

Sumin Kim

Ph.D. Student at Seoul National University

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Research Interest

Video Understanding, Multimodal Learning, Representation Learning, Computer Vision, Deep Learning

Education

Seoul National University, Seoul, South Korea

Mar 2025 – Present

Ph.D. Student in Data Science

- Advisor: Prof. Joonseok Lee ([link](#))
- Visual Information Processing Lab. ([link](#))

Sungkyunkwan University, Seoul, South Korea

Mar 2020 – Feb 2025

B.S. in Applied Artificial Intelligence

- GPA: 4.23/4.50 (Magna Cum Laude)

Publications

*: Equal Contribution †: Corresponding Author

C3 From Matching to Reasoning: Query-Aware Long Video Summarization

Mingu Kang, **Sumin Kim**, Hyunjin Lee, Sungmin Yang, Yoori Oh[†], Joonseok Lee[†]
Conference on Neural Information Processing Systems (NeurIPS), 2026

C2 TripleSumm: Adaptive Triple-Modality Fusion for Video Summarization

Sumin Kim^{*}, Hyemin Jeong^{*}, Mingu Kang^{*}, Yejin Kim, Yoori Oh[†], Joonseok Lee[†]
International Conference on Learning Representations (ICLR), 2026
[Paper](#) | [arXiv](#) | [Website](#) | [Code](#) | [Dataset](#)

C1 SummDiff: Generative Modeling of Video Summarization with Diffusion

Kwanseok Kim^{*}, Jaehoon Hahm^{*}, **Sumin Kim**, Jinhwan Sul, Byunghak Kim, Joonseok Lee[†]
International Conference on Computer Vision (ICCV), 2025
Highlight Presentation (263/11,239 = 2.34%)
[Paper](#) | [arXiv](#) | [Website](#) | [Code](#)

Patents

Video Summarization Method and Computer Apparatus for the Same

May 2026

Korean patent application filed May 24, 2026; application number pending

Inventors: **Sumin Kim**, Hyemin Jeong, Mingu Kang, Yejin Kim, Yoori Oh, and Joonseok Lee

Method for Generating Video Summaries and Computer Apparatus Therefor

Apr 2026

Korean Patent Application No. 10-2026-0063327, filed Apr. 8, 2026

Inventors: Kwanseok Kim, Jaehoon Hahm, **Sumin Kim**, Jinhwan Sul, Byunghak Kim, and Joonseok Lee

Work Experience

Undergraduate Researcher, Korea Institute of Science and Technology

Feb 2023 – Jun 2023

Center for Artificial Intelligence, AI & Robotics Institute

Seoul, South Korea

- Investigated parameter-efficient adaptation for video understanding, balancing efficiency and model performance.

Projects

- | | |
|---|---------------------|
| Multi-modal Fusion for Video Summarization @NRF Grant | Sep 2025 – Aug 2026 |
| <ul style="list-style-type: none">Selected for the 2025 Basic Science Research Program funded by the National Research Foundation of Korea. Leading a research project to develop a robust video summarization framework by fusing multimodal data. | |
| Generative AI for Visual Effects @Samsung Electronics | Jun 2025 – Dec 2025 |
| <ul style="list-style-type: none">Designed an SDXL-based visual-effects generation pipeline using LoRA fine-tuning, balancing style transfer strength with object-level semantic and geometric preservation. | |

Fellowships and Scholarships

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| NRF Graduate Research Encouragement Grant
<i>National Research Foundation of Korea (NRF)</i> | Sep 2025 – Aug 2026 |
| Yulchon AI Research Fellowship
<i>Yulchon Foundation</i> | Aug 2025 |
| Presidential Science Scholarship, Engineering
<i>Korea Student Aid Foundation</i>
National merit-based scholarship; selected as one of 60 national scholars | Mar 2023 – Feb 2025 |
| Student Success Leader Scholarship
<i>Sungkyunkwan University</i> | 2022, 2023, 2024 |
| Excellence Performance Scholarship
<i>AI Convergence and Open Sharing System</i> | Jan 2024 |
| University Innovation Scholarship
<i>Sungkyunkwan University</i> | 2021, 2023 |
| Academic Excellence Scholarship
<i>College of Computing and Informatics, Sungkyunkwan University</i> | Dec 2023 |

Honors and Awards

- | | |
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| Excellence Award, Industry-Academic-Linked AI Hackathon
<i>AI Convergence and Open Sharing System</i> | Aug 2024 |
| Dean's List Award, High Scholastic Achievement
<i>College of Computing and Informatics, Sungkyunkwan University</i> | Dec 2023 |
| SK Telecom Special Award, Seoul Central AI Competition
<i>Sungkyunkwan Univ., Hanyang Univ., Kyung Hee Univ., Univ. of Seoul, Konkuk Univ., Sejong Univ., Inha Univ., Catholic Univ.</i> | Nov 2023 |
| Special Award, Uni-DTHON (AI & Data Track)
<i>Seoul Nat'l Univ., KAIST, POSTECH, Yonsei Univ., Korea Univ., Sungkyunkwan Univ., Chung-Ang Univ., Sookmyung Women's Univ.</i> | Nov 2023 |
| Grand Award, Capstone Design and AI Hackathon Competition
<i>The Korean Association of Computer Education</i> | Oct 2023 |
| Excellence Award, KOPIS Big Data Analysis Contest
<i>Ministry of Culture, Sports and Tourism</i> | Sep 2023 |
| Excellence Award, AI Service Design Hackathon
<i>Samsung Electronics</i> | Aug 2023 |
| Top Award, Co-Deep Learning Project Competition
<i>Sungkyunkwan University</i> | Jan 2023 |
| President's List Award, University-wide Undergraduate Honor
<i>Sungkyunkwan University</i>
Selected as one of the top 20 students university-wide | Dec 2022 |
| Excellence Award, Web Development Competition Using AI
<i>College of Computing and Informatics, Sungkyunkwan University</i> | Aug 2022 |
| Top Award, Web Development Competition Using AI
<i>College of Computing and Informatics, Sungkyunkwan University</i> | Aug 2021 |

Invited Talks

“Practical Workshop on AI Agent-Based SDD (40 Hours)” , Hanwha Systems	Aug 2026
“Where Do Good Research Ideas Come From?” , Sungkyunkwan University	May 2026
“Why Do We Pursue Graduate Studies?” , Yonsei University	Mar 2026
“Multimodal Video Summarization Research” , Korea University	Sep 2025
“Reflections on Video Research” , Sungkyunkwan University	May 2025

Teaching

Teaching Assistant , Core Computing: Thinking with Computers – SNU	Fall 2026
Teaching Assistant , Machine Learning for Visual Understanding – SNU	Spring 2026
Teaching Assistant , Core Computing: Thinking with Computers – SNU	Spring 2026
Teaching Assistant , Programming for Data Analysis – SNU	Winter 2026
Teaching Assistant , Core Computing: Thinking with Computers – SNU	Fall 2025
Teaching Assistant , Basic Computing: First Adventures in Computing – SNU	Summer 2025
Teaching Assistant , Computational Thinking and Software Coding – SKKU	Spring 2024
Teaching Assistant , Artificial Intelligence Basics & Uses – SKKU	Fall 2022

Academic Services

Reviewer

- International Conference on Learning Representations (ICLR) 2027
- Conference on Neural Information Processing Systems (NeurIPS) 2026
- International Conference on Machine Learning (ICML) 2026

Activities

Founding President , AI Academic Student Society “brAIIn” – SKKU	Sep 2024 – Feb 2025
President , School of Global Convergence Student Council “cresc.” – SKKU	Nov 2021 – Nov 2022
Member , Central Student Steering Committee – SKKU	Nov 2021 – Nov 2022
President , School of Global Convergence Ambassadors “Alimi” – SKKU	Sep 2021 – Aug 2022
Visiting Student , Intensive Programming and Startup (CS496) – KAIST	Dec 2021 – Feb 2022